

MINI-PROJECT SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Mini-Project Title: Mini-Project 2 — Week 5 Capacitor (Trovey Native Shell)
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Submission Date: 10/09/2026

1. GENERAL INFORMATION & DELIVERABLE LINKS

Team Members:

1. Nguyễn Minh Duy — Student ID: 23IT038 — Role: Solo (PWA HTML/JS, Capacitor Android, sync, deploy) — Contribution: 100%

Live Demo URL: <https://pwa.puretrovey.net/>
Flutter PWA (branch): <https://app.puretrovey.net/> git checkout mob-flutter
GitHub Repository: <https://github.com/minhduy6868/MOB-Survey>
Android APK: <https://pwa.puretrovey.net/downloads/trovey.apk>
Video Demo: Not submitted.

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required Feature	Status	Implementation Details & Acceptance Level
1	Capacitor + Android platform	Complete	<code>npx cap add android</code> . Project android/. capacitor.config.json webDir = pwa/. Appld net.puretrovey.trovey.
2	Camera plugin replaces file input	Complete	<code>pwa/native.js</code> <code>troveyTakePhoto: Camera.getPhoto({ resultType: base64, source: CAMERA })</code> when <code>Capacitor.isNativePlatform()</code> ; else <code><input capture></code> .
3	Geolocation plugin	Complete	<code>troveyGetGps</code> uses <code>@capacitor/geolocation</code> on native; <code>navigator.geolocation</code> on the browser PWA.
4	Filesystem before sync	Complete	After camera, <code>Filesystem.writeFile</code> to <code>DATA/photos/field-{ts}.jpg</code> . New capability vs Week 3.
5	Sync-success notification	Complete	Lab asked Push. Repo has no <code>google-services.json</code> / FCM. Local Notifications fire when status becomes synced. Same user goal, app need not stay in a browser tab.
6	Build APK	Complete	<code>npx cap sync && gradle assembleDebug</code> . File <code>pwa/downloads/trovey.apk</code> . Trovey clipboard launcher + splash (not the default Capacitor mark).
7	Device test path	Complete	Permissions screen on first launch (Camera, GPS, Notifications, Files). Form still 3 steps. Queue <code>draft queued syncing synced failed</code> unchanged.
8	Do not rewrite the app	Complete	Sync engine, dynamic form, IndexedDB (<code>trovey-pwa / tickets</code>) are the Week 3/HTML PWA. Capacitor only wraps them.

3. TECHNICAL ARCHITECTURE & PROJECT STRUCTURE

The web app (HTML/CSS/JS in pwa/) runs unchanged in the Android WebView. pwa/native.js is the Capacitor bridge: it calls Camera, Geolocation, Filesystem, and LocalNotifications only when Capacitor.isNativePlatform() is true. The browser PWA keeps file input, navigator.geolocation, and Notification.

Directory map:

pwa/	Web app copied into the native shell (webDir)
pwa/native.js	JS ↔ native plugins (camera, GPS, files, notify)
pwa/app.js	Form, IndexedDB queue, results charts, drain()
android/	Editable Android Studio project (Capacitor 7)
android/.../AndroidManifest.xml	CAMERA, LOCATION, POST_NOTIFICATIONS
functions/api/sync.js	POST /api/sync → Cloudflare KV (unchanged)
functions/api/records.js	GET /api/records JSON/CSV (unchanged)
capacitor.config.json	appld net.puretrovey.trovey, webDir pwa

State flow: form edit → IndexedDB draft → Submit → status queued → POST

<https://app.puretrovey.net/api/sync> → KV put ticket:{clientId} → status synced → Local Notification (native) or Web Notification (browser). clientId is a UUID, so retry does not create a second row. On online, pullCloud() merges remote rows, then drain() sends queued/failed.

Exception handling: HTTP or body ok!=true / kv!=true throws. sendOne() sets failed + lastError. pullCloud() swallows network errors. A ticket is accepted when KV write succeeds. Sheet webhook 401 does not fail the ticket (same as Week 3).

What the Week 5 slides said must not change: sync engine, dynamic form, IndexedDB. Those live in pwa/app.js + functions/api/. Week 5 changed the three hardware APIs plus a local photo write, and wrapped the PWA in Capacitor.

4. EMPIRICAL EVIDENCE & SCREENSHOTS

Screenshots from <https://pwa.puretrovey.net/> on a 390×844 viewport (Chromium). Captured 10/09/2026. Four field interviews were filled on the same web form the APK embeds (three steps, skip GPS like a rushed site visit) and synced with kv:true:

- Phạm Thị Hạnh — Đà Nẵng, quán, 25–34, cổ phiếu+forex, swing, VNDirect, stop-loss có, lãi nhẹ.
- Trần Quốc Bảo — TP.HCM, văn phòng, 35–44, futures+chỉ số, day, DNSE, không stop, thua 3 tháng.
- Lê Minh Châu — Huế, ký túc, 18–24, crypto, copy, Binance, stop-loss có, hòa vốn.
- Nguyễn Hoàng Nam — Hà Nội, gọi từ xa, 25–34, forex+hàng hóa, position, MetaTrader 5, lãi rõ.

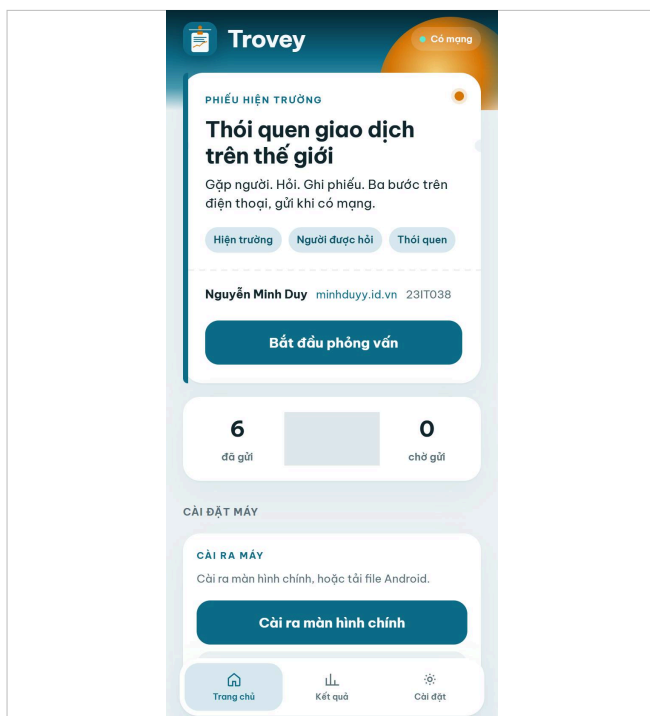


Figure 1. Home: topic, start CTA, 6 submitted / 0 queued, install + APK.

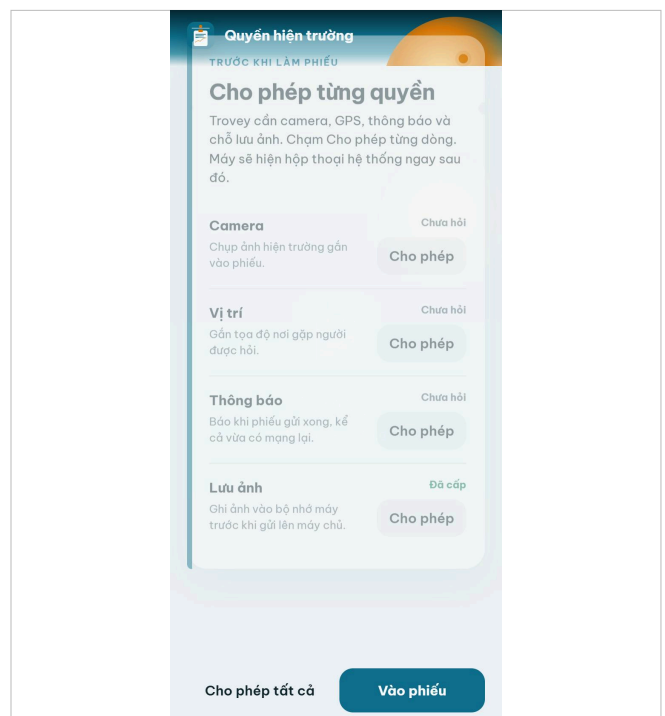


Figure 2. First-launch permissions (Camera, GPS, Notifications, Files) before any ticket.

Figure 3. Form step 1 (site): Việt Nam, Đà Nẵng, quán — GPS skipped. Draft saved on device.

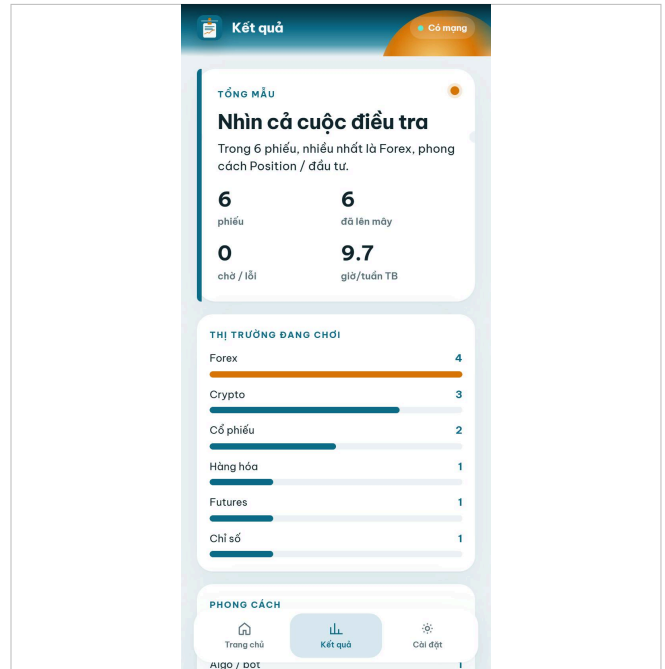


Figure 4. Results overview: sample KPIs, market bars, style chart — not a flat list.

Figure 5. Settings: collector 23IT038, shell label (PWA trình duyệt vs vỏ native), APK download.



Figure 6. Android round launcher after branding (Trovey clipboard, not Capacitor default).

5. TECHNICAL CHALLENGES & RESOLUTIONS

5.1 Capacitor 7 needs JDK 21+

assembleDebug with JDK 17 failed: invalid source release: 21 (capacitor.build.gradle compileOptions VERSION_21). Resolution: build with JDK 22 (JAVA_HOME). README states JDK 17 is not enough.

5.2 APK launcher showed the Capacitor mark

Legacy ic_launcher.png was Trovey, but Android 8+ adaptive icons use mipmap foreground + splash drawables that were still the default Capacitor logo. Resolution: logos/apply_android_brand.py rasterizes pwa/icons/Icon-512.png and Icon-maskable-512.png into all mipmap densities and splash sizes; ic_launcher_background = sampled teal.

5.3 Push Notifications vs Local Notifications

The slide asks @capacitor/push-notifications so the alert works without the browser staying open. FCM needs google-services.json, which is not in the repo. Resolution: @capacitor/local-notifications on native (and Web Notification on the PWA) when a ticket becomes synced. Same lab goal; no Firebase project.

5.4 Cloudflare Pages 25 MiB file limit on the APK

A debug APK that had already been copied into pwa/downloads/ was then packed again into android assets (webDir = pwa), so the next APK grew past 25 MiB and Pages rejected downloads/trovev.apk. Resolution: ignore *.apk in aapt; scripts/build-apk.cjs deletes the web APK before cap sync, then copies the new APK after assembleDebug. Upload is 17.5 MiB at <https://pwa.puretrovey.net/downloads/trovev.apk>.